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MILITARY SPECIFICATION

INDICATOR, ATTITUDE REFERENCE SELF-CONTAINED, 3-INCH, ID-1791/A

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - The equipment covered by this specification is a Self-Contained Vertical Gyro Indicator System for Military Aircraft.

2. APPLICABLE DOCUMENTS

*2.1 General - The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-S-8802	Sealing Compound, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 6610

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Military (continued)

MIL-M-7793	Meter, Time Totalizing
MIL-A-8623 or MMM-A-134	Adhesive, Epoxy Resin, Metal to Metal Structural Bonding
MIL-A-9067	Adhesive Bonding, Process and Inspection Requirements for
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-T-18303	Test Procedures, Preparation and Acceptance for Electronics Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-I-81219	Indicator, Elapsed Time, Electrochemical
MIL-G-81322	Grease Lubrication for Ball Bearings
MIL-C-81511	Connector, Electric, Circular, High Density, Quick Disconnect, Environment Resisting
MIL-G-81559	Gyroscope Assemblies, Attitude and Directional Reference Instruments for Aircraft, Preservation and Packaging, Procedures for
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover
MIL-G-81744	Barrier Coating and Solution
MIL-B-81793	Bearing, Ball, Precision, for Instruments and Rotating Components
MIL-L-81846	Lubricating Oil, Instrument, Ball Bearing, High Flash Point
MIL-G-81937	Grease, Instrument Ultra Clean

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STANDARDS

Federal

FED-STD-209	Clean Room and Work Station Requirements, Controlled Environment
FED-STD-595	Colors
DD-G-451	Glass, Flat and Corrugated for Glazing Mirrors and other uses

Naval Air Systems Command

AR-5	Microelectronic Devices used in Avionics Equipment Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

Military

MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-471	Maintainability Demonstration
MIL-STD-480	Configuration Control-Engineering Changes and Waivers
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-1334	Barrier Coating Process for Anti-Friction Bearings

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Military (continued)

MIL-STD-2074	Failure Classification for Reliability Testing, General Requirements for
MS-24367	Lamp Incandescent, Miniature Integral Lighting
MS-33558	Numerals and Letters, Aircraft Instrument, Dial, Standard Form of

2.2 Availability of Documents - When requesting specification, standards, drawings, and publications, refer to both title and number. Copies of this specification and required specifications by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 First Article (Preproduction) - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies shall be used in accordance with Specification AR-10 and as outlined in Specification MIL-E-5400.

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(4) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors, and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

*3.2.1.1 Final Approval - Approval of all nonstandard parts and materials shall be completed before First Article approval can be granted.

3.2.2 Nonmagentic Materials - Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.2.3 Aluminum Alloy Parts - Aluminum alloy parts shall be covered with an anodic film conforming to MIL-E-5400 except that MIL-STD-171 may be used when required.

3.2.4 Microelectronic Modular Assemblies - When used, Microelectronic Modular Asemblies shall meet the requirements of Specification AR-5. Conformal coatings encapsulants, embedments, or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.5 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10.

*3.2.6 Configuration Management - Upon approval of First Article (Preproduction) testing, any or all alterations to the design, configuration, or part changes shall be in accordance with MIL-STD-480.

*3.3 Design and Construction - The indicator shall consist of an indicator case containing a vertical gyro, a sphere to indicate pitch and roll, a bank index to indicate roll, related circuitry and mechanism, a power failure indicating device, a mechanism to adjust

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pitch trim, and a fast erect capability. Pitch and roll pickoffs shall develop electrical signals representative of aircraft attitude. A turn rate display for control by remote circuit, an inclinometer and integral lighting shall be provided. Cross pointers on the indicator presentation shall also be provided. The indicator shall be so designed and constructed that adjustment and repairs can be easily made by the personnel of operating units and overhaul bases. It shall withstand the temperature extremes, the strains, jars, and vibrations incident to shipping, storage, installation, and service. The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction, and workmanship, except as otherwise specified herein. Cantilever mounting of gyro and sphere from the rear end of the case should be avoided. The drive to the sphere shall not permit an oscillation or jitter greater than ± 0.3 degree.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 4.75 pounds (2.155 kgs).

3.3.2 Reliability - The supplier shall conduct a reliability program using MIL-STD-785 as a guide.

*3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 2000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum operating life of 5,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The indicator shall have a mean time between failure of 2,000 hours when tested and accepted as outlined under the requirements of 4.4.3.

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3.3.2.4 Time Totalizing Meter - Elapsed time meter MIL-M-7793 (5000 hours) shall be used unless space considerations require alternate MIL-I-81219. The reading of the meter shall not require unsealing of the indicator case. The time totalizer shall be installed within the envelope of the instrument.

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment. Flat flexible ribbon cabling shall not be used without specific approval of the procuring agency.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-STD-461.

*3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400, and AR-10.

3.3.7 Nomenclature, Nameplates, and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

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*3.3.7.1 Bearing Barrier Coating - The instrument shall be identified as having bearings coated in accordance with MIL-STD-1334. The tag shall conform with Specification MIL-N-18307. Interfacing parts need not be processed where ball bearings have chamfered races barrier coated unless tests indicate otherwise. Each unit shall be identified by a red background decal with white letters as follows: "Barrier Coated Per MIL-STD-1334."

*3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room Ambient (25°C ± 10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115/200 VRMS ± 1% 3 phase, 400 Hz and 5.0 ± .1V AC/DC

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein.

*3.3.10 Vibration - The indicator shall be capable of operating under all vibratory conditions as specified herein.

*3.3.11 Warm Up Time - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 10 minutes under extreme service conditions.

*3.3.12 Insulation Resistance - With a potential of 500 volts direct current applied for 5 seconds between isolated circuits and between circuits and the case, the resistance shall be greater than 50 megohms.

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3.3.13 Input Electrical Power -

*3.3.13.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

(1) AC power, 400 Hz, three phase, 115/200V line-to-line Category B, 19 VA MAX, running.

3.3.13.2 Lighting Power - The integral lighting shall use 5V AC or DC, 5VA maximum power derived from the power supply normally used for lighting the other instruments. (see 3.5.10)

3.3.13.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electronic system operation. Operation shall return to normal with no resulting damage to the equipment.

*3.3.14 Explosive Atmosphere - The indicator, when operated in an explosive atmosphere, shall not cause an explosion.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The indicator shall provide accurate visual presentation of aircraft displacement in the roll and pitch axes which are referenced to gravity, without requiring inputs from any external source other than the power supply. The rate-of-turn pointer shall provide accurate visual presentation

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when properly energized from an external source. The pitch and roll pick-offs shall provide electrical information through 360 degrees in roll and approximately +92° -78° in pitch of aircraft attitude. The pitch synchro output shall not show effects of cross coupling, due to aircraft roll maneuvers, greater than plus one minus zero degree for roll attitudes up to 30°.

3.4.1.1 Accuracy -

3.4.1.1.1 Standard Conditions - The indicator, when connected and operated at standard conditions, shall provide a pitch and roll accuracy as follows:

Visual Presentation:

Roll 0.50° Maximum Error
Pitch 0.50° Maximum Error

Pick-Off:

Roll 0.50° Maximum Error Output
Pitch 0.50° Maximum Error Output

3.4.1.1.2 Longitudinal Acceleration - The indicator, when subjected to an 8g acceleration for 4 seconds along the roll axis, shall provide a pitch and roll accuracy as follows:

Visual Presentation:

Roll 2.0° Maximum Error
Pitch 2.0° Maximum Error

Pick-Off:

Roll 2.0° Maximum Error Output
Pitch 2.0° Maximum Error Output

3.4.1.1.3 Coordinated Turns - The indicator, when subjected to coordinated turns with rates of turn between 90 and 1500 degrees per minute and bank angles of 20 to 60 degrees, shall provide a pitch and roll accuracy as follows:

Visual Presentation:

Roll 2.0° Maximum Error
Pitch 2.0° Maximum Error

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Pick-Off:

Roll 2.0° Maximum Error Output
Pitch 2.0° Maximum Error Output

3.4.1.2 Performance at Environmental Extremes - Performance deviations at environmental extremes shall be permitted as specified in the detail requirements.

3.5 Detail Requirements -

3.5.1 Vertical Gyroscope - The indicator shall comprise a 115V/200V 3 phase 400 cycle operated vertical gyroscope mounted in a case with its outer gimbal axis fore and aft, and its inner gimbal axis athwartship. The vertical gyroscope shall be referenced to gravity with compensation for acceleration resulting from turns. A moveable roll index shall move over a scale fixed to the case. A moveable sphere shall register movement in the pitch and roll axes with respect to a miniature airplane mounted on an adjustable bracket and located at the center of the indicator glass cover. The attitude presentation shall be from 0 to 360° roll, and 92° climb, 78° dive. The indicator shall be adjusted to indicate zero pitch when mounted with the case tilted 7° from the horizon with the connector end down. Rotation of a knob located on the bezel, shall adjust the position of the miniature airplane and the horizontal flight director pointer to a minimum of 10° in dive and 5° in climb. Pulling the knob shall cage the gyro in pitch and roll. Rotating the knob clockwise while it is fully out shall lock the gyro in the caged position. A warning flag shall come into view when the gyro is caged, the AC power supply is interrupted, or the phase rotation is reversed. The gimbal pick-offs shall develop electrical signals representative of indicator attitude through 360° roll and approximately 92° climb, 78° dive in pitch.

3.5.1.1 Angular Momentum - The gyro shall be of such design and shall possess sufficient angular momentum to insure that the performance required in this equipment will be met throughout all operating conditions specified herein.

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3.5.1.2 Spin Axis Verticality - The erection system shall be capable of maintaining the gyro spin axis within 1/2° of vertical, within 3.5 minutes after application of power.

*3.5.1.3 Gimbal Freedom - Unlimited freedom shall be provided about the roll axis. Pitch stops shall be incorporated which produce a roll precession when pitch freedom is exceeded. During power interruption, normal freedom is maintained in both axes. The pitch stop shall be so designed that no parts of the instrument will be damaged by a loop at the rate of 90° per second and less than 3° error will result from loops of 12 ± 1° per second. Errors induced by roll maneuvers at the rate of 300° per second shall not exceed 1°.

*3.5.1.4 Caging - Means shall be incorporated to cage the gyroscope by pulling the pitch trim knob. When the knob is pulled and power is applied, the power failure flag shall appear and continuity between pins 20 and 21 of the connector shall be closed.

*3.5.1.5 Erection - An erection system which will maintain the alignment of the spin axis to the local vertical shall be provided. Roll erection shall be cut off automatically when a false vertical in excess of 5° ± 3/4° results from a turn. Erection shall be obtained without vibration (scorsby motion may be used).

*3.5.1.5.1 Initial Erection - The caging knob shall erect the gyro to within 0° ± 1° Roll and 4° ± 1° Climb from any position at any time after power is applied.

*3.5.1.5.2 Final Erection - The gyro spin axis shall erect to within 0.5° of vertical within 3.5 minutes after release of the caging knob.

*3.5.1.5.3 Erection Rate - The gyro shall be capable of erecting in pitch and roll at a rate of 1.3 to 2.5° per minute. Roll erection shall be cut off when the spin axis displacement exceeds 5 ± 3/4° in roll.

*3.5.1.6 Coast Time - The indicator shall display pitch and roll information within 6° of vertical for a minimum of 3 minutes after power is removed.

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3.5.2 Markings and Lines - All markings of the indicator display shall be clean and sharp. Dimensions shall be as specified. On any given mark, the width shall not vary noticeably throughout the length of the mark. All markings which are adjacent for some specific indication for the same basic dimension shall not be noticeably different. Unless otherwise specified, the dimensions of all markings on the indicator shall be in accordance with Figures 2, 3, 4, 5, 6, and 7 and Table I.

3.5.3 Numerals and Letters - All numerals and letters on the indicator shall be in accordance with MS33558.

3.5.4 Pitch and Roll Display - The display shall be in accordance with Figure 1, and shall consist basically of an adjustable miniature airplane, a bank angle dial, a bank index, and a moving two-colored spheroid background with a distinct horizon line dividing the two colors. Unless otherwise specified, all colors shall be in accordance with FED-STD-595, with all visible portions of the indicator case being black color 37038. The paint shall diffuse light sufficiently to avoid "high light" when illuminated with external or internal light sources. The bank angle dial, the bank index, and the miniature airplane shall be designed to minimize parallax effects when interpreting the display.

*3.5.4.1 Adjustable Miniature Airplane - The design shall be in accordance with Figure 2. The rear edge of the miniature airplane shall be within .090 inch of the spheroid surface. The wings shall follow the curvature of the spheroid. The top edge of the wings and the center dot may be tilted back approximately 45° to facilitate both integral and external lighting. The miniature airplane shall be adjustable, having a range of +5° -10°, from the position shown in Figure 1. Adjustment shall be made with a knob on the front plate (see Figure 1). A clockwise motion of the knob shall move the airplane to indicate climb and a counterclockwise motion will move the airplane to indicate dive. The miniature airplane shall have a black border of uniform width of .010 ± .005 inch around the wings and center dot. The painted portion of the "W", the wings, and the center dot shall be separated by an area of sufficient width with black color 37038 to give an area of contrast.

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3.5.4.1 Pitch Trim Knob - Means shall be incorporated to adjust the position of the miniature airplane by means of a pitch trim knob mounted on the lower right hand side of the face of the indicator and marked in accordance with Figure 1. The reference pointer on the knob shall align with the index scale on the dial at the zero pitch trim position. The body color of the knob shall be black color 37038 of FED-STD-595, and the reference index scale on the dial shall be white color 37875 of FED-STD-595. The freedom in each direction shall be limited by a positive stop. When the knob is rotated clockwise to the stop, the miniature airplane shall move upward 5°. When the knob is rotated counterclockwise to the stop, the miniature airplane shall move downward 10°. The knob shall have sufficient friction to remain fixed when set but it shall move smoothly without evidence of binding or erratic motion. The pitch trim shaft shall be adequately sealed to prevent salt water penetration.

3.5.4.3 Bank Index - The bank index shall be in accordance with Figure 3. It shall be affixed to rotate with the roll gimbal. The index shall not move with pitch deviations. The bank index shall be with 1/32 ± 1/64 inch of the small diameter edge of the scale surface of the bank angle dial. At zero roll indication, the bank index shall be aligned with the zero marking on the bank angle dial. The index shall be white color 37875 of FED-STD-595. The border shall be black color 37038 of FED-STD-595.

3.5.4.4 Bank Angle Dial - The bank angle dial shall be in accordance with Figure 5. The scale markings of the dial shall be white color 37875 of FED-STD-595. The background shall be black color 37038 of FED-STD-595.

3.5.4.5 Spheroid - The spheroid shall rotate about the roll axis of the case for roll indications and shall rotate about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator. The pitch axis shall be aligned with the center of the display when the indicator is indicating a zero roll attitude. The spheroid shall be driven in pitch in such a way that for a dive maneuver, the horizon line goes above the miniature airplane. The spheroid shall be in accordance with Figures 5 and 6, the surface of the spheroid shall move .033 (nominal) inch for each degree of pitch change.

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3.5.4.5.1 Spheroid Markings and Colors - Above the horizon line, the spheroid shall be light gray color 36440 of FED-STD-595 and all markings shall be black color 37038 of FED-STD-595. The horizon line shall be white color 37875 of FED-STD-595. Below the horizon line the spheriod shall be black color 37038 of FED-STD-595 and all markings shall be white 37875 of FED-STD-595. The lengths of the major pitch lines shall be in accordance with the following table. The width of lines shall be in accordance with Figure 6.

0°	(horizon line)	Width of Spheroid
5°		.22 inch
10°		1.69 inch
15°		.22 inch
20°		1.53 inch
25°		.22 inch
30°		1.34 inch
35°		.22 inch
40°		1.16 inch
45°		.22 inch
50°		.97 inch
55°		.22 inch
60°		.78 inch
65°		.22 inch
70°		.59 inch
75°		.22 inch
80°	Climb	.41 inch
87°	Climb Circle	.66 inch dia
92°	Climb .Solid Dot	.33 inch dia
73°	Dive Circle	.66 inch dia
78°	Dive .Solid Dot	.33 inch dia

The center of the solid dots shall be the extreme pitch position and from center to center shall represent 180 degrees pitch indication.

3.5.5 Inclinometer - The inclinometer shall be incorporated into the display in accordance with Figures 1 and 7.

3.5.5.1 Inclinometer Tube Glass - The inclinometer shall be made of clear annealed glass tubing free from flaws that will affect the readability. The inside of the tube shall be smooth. The tube shall be painted per Table I. The wall thickness shall not be less than 3/64 inch.

3.5.5.2 Inclinometer Ball - The ball shall be an accurate sphere, 5/32 ± 1/64 inch in diameter, of black glass color 17038 of FED-STD-595. The color of the lubber wires on the slip

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indicator shall be the same as the slip indicator ball. a minimum of two-thirds of the ball shall remain in view at either extreme of the tube when viewed from a position 12 inches in front of the zero mark of the slip indicator at an angle of 30° with the horizontal plane prescribed by the base of the indicator.

3.5.5.3 Damping Liquid - The damping liquid shall be sufficiently colorless under all specified conditions to preclude interference with visibility of the ball. The viscosity shall be such that with the ball in motion at the zero mark, and the case tilted 12°, the time required for the ball to roll from the zero mark to the end of the tube shall be not less than 0.5 second.

3.5.5.4 Sensitivity - The sensitivity shall be such that the ball reaches its limit of travel when the indicator is rotated right or left 10 ± 2° about the roll axis.

3.5.5.5 Inclinometer Filling - No part of the air bubble shall be visible when the indicator is viewed from the position specified in 3.5.5.2.

3.5.6 Rate-of-Turn - The rate-of-turn display shall be in accordance with Figures 1 and 7. The design shall be such that the pointer is stable as practicable under rough air conditions and 1/2g vibration. Within normal operating position, the index shall move left or right in agreement with the polarity of the direct current received from the remote rate-of-turn sensor. The scale shall conform to the path of the index so that there is a minimum of parallax. The index deflection to either side of the center zero position shall be limited to 1/2 inch. At each of the fully deflected positions, a minimum of one-half of the index shall be in view when viewed at a horizontal distance of 2 feet from the center point of the display. Fixed mechanical stops shall be provided at the extreme deflection points to prevent pointer hangup on the case or mechanisms.

3.5.6.1 Sensitivity Element - The sensitive element of the turn indicators shall be operated on 1 vdc (2 vdc maximum input) applied to pins 26 and 27. The element shall be so mounted that it reacts only to turns about the vertical axis.

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3.5.6.2 Pointer - The pointer shall be as shown on Figure 7.

*3.5.6.3 Scale - The turn rate scale shall conform to Figure 7. The colors shall be as specified in Table II. The distance between the indicator dial and the inside of the cover glass shall be held to a minimum and shall not exceed 1/4 inch at the zero position.

TABLE I. Inclinometer Tube Color.

Marking	Color
Visible Part of Two Wires on Front and Top Inclinometer	Black No. 37038
Wire on each side of Inclinometer	Black No. 37038
Inclinometer Background	White No. 37875
All Other Background	Black No. 37038

TABLE II. Turn Rate Scale Color.

Marking	Color
Center Mark	White No. 37875
Two Needle-width Deflection Marks	White No. 37875
Pointer	White No. 37875

*3.5.6.4 Deflection (Pointer) - The deflection for a given input shall be as specified in the following table. The maximum voltage supplied to pins 26 and 27 shall be limited to 2 vdc. For left turns (cw deflection), the voltage polarity shall be positive (+) on pin 26 and negative (-) on pin 27. For right turns (ccw deflection), the voltage polarity shall be negative (-) on pin 26 and positive (+) on pin 27. Over travel shall be 1/2 needle width minimum.

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Rate-of-Turn (Degrees/Min)	Input Current (Microamperes)	Deflection (Inches)
0	0 ± 7.5	0
90	500 ± 38	.156
180	1000 ± 65	.312

*3.5.6.5 Response Time - The response time (see 6.5.1) of the rate-of-turn index shall be a maximum of .45 sec.

3.5.7 Flight Director Display - The flight director display shall consist of a vertical pointer and a horizontal pointer. These pointers shall be driven by meter movements and shall be in accordance with Figures 1 and 4. The horizontal pointer shall be nearest the indicator glass. The design of the pointer meter movements shall be such that when energized the pointers will be maintained as near as practical to their adjusted zero positions, while under all flight conditions, including rough air. When the meter movements are not used, the pointers shall be driven out of view. Out-of-view position for the vertical pointer shall be at the right side of the display. Out-of-view position for the horizontal pointer shall be at the bottom of the display. All meter movement leads shall be electrically isolated from each other, from the circuitry in the case, and from the case. Unless otherwise specified, all meter movements shall operate as specified herein when operated singly, when connected in parallel with a like meter movement, or when connected parallel with a 1,000 ohm load to a signal source that has an impedance between 350 and 1,000 ohms. The yellow area of the horizontal and vertical steering pointers shall be centered over the tail of the airplane. External mechanical zeroing adjustments shall be provided where required.

3.5.7.1 Vertical Pointer - The vertical pointer shall be positioned behind the indicator light wedge in front of the miniature airplane. The vertical pointer shall be mounted on a meter movement that is pivoted at a point approximately 3-3/32 inches behind the pointer in such a manner that the pointer will remain vertical at zero center or at any point of deflection to the right or left of center, and will deflect right or left in agreement with the polarity of direct current received from the flight director computer. The portion of the vertical pointer dimensioned in Figure 4 shall be color 33695 of FED-STD-595.

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*3.5.7.1.1 Deflection - The design of the vertical pointer shall be such that the deflection of the pointer to the left of the center position will be limited to 5/8 inch by a physical stop, and the maximum deflection to the right will carry the pointer behind the mask on the right side of the display. The direction of the deflection for a given input signal shall be as specified in Table III.

*3.5.7.1.2 Response - The vertical pointer response shall be linear in degree with respect to current within 7.5 percent of the proportional full scale value. A deflection of 1/2 inch, measured from the center of the dial to the pointer along the horizontal line shall require 2.2 ma ± 7.5 percent; 2.5 ma shall not deflect the pointer by more than 5/8 inches. An application of 10 ma of the pointer polarity (as shown in Table III) shall drive the pointer from view to behind the shield at the right. The application of 13.5 ma as a step input to drive the pointer from view shall not damage the meter movement.

3.5.7.1.3 Resistance - The resistance across the vertical pointer input terminals shall be 1,000 ohms ± 3 percent at 25 degrees centigrade.

3.5.7.1.4 Damping - The pointer mechanism shall be so damped that there will be no more than 1-1/2 percent overshoot.

*3.5.7.1.5 Response Time - Response time of the vertical pointer shall be a maximum of .45 second. Response time is the time required for the pointer to reach 90 percent of its final indication time shall be counted from the instant of application of current. The pointer shall be considered as having come to apparent rest when it has reached within ± 1 percent of the scale length of the actual rest point.

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TABLE III. Pins Designation.

Pin	Function
1	115/200V, 400 Hz, Phase A NOTE: Phase rotation A, B, & C.
2	115/200V, 400 Hz, Phase B
3	115/200V, 400 Hz, Phase C
4	Internal Lighting 5V AC or DC (Red Lighting)
5	Internal Lighting, Ground
6	+ Vertical Pointer (Deflects to right)
7	- Vertical pointer " "
8	+ Horizontal Pointer (Deflects downward)
9	- Horizontal Pointer " "
10	Neutral Power
11	Ground Flight Director Mode Engagement
12	28VDC Flight Director Mode Engagement
13	(X)
14	(Y) Roll Synchro Output
15	(Z)
16	(X)
17	(Y) Pitch Synchro Output
18	(Z)
19	Case Ground
20	}
21	} Normally closed when indicator caged.
22	Spare
23	Spare
24	Spare
25	Spare
26	(+) Rate-of-Turn Input (left turn) } 1000 ohm normal load
27	(-) Rate-of-Turn Input " " }
28	}
	} Jumper pins 28 and 29 for 500 ohm rate input load.
29	}
30	Spare
31	Spare
32	Reserved
33	Reserved
34	Reserved
35	Spare
36	Spare
37	Spare

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*3.5.7.2 Horizontal Pointer - The horizontal pointer shall be positioned in front of and as near as practicable to the vertical pointer. The horizontal pointer shall be mounted on a meter movement which shall be pivoted at a point approximately 3-1/32 inches behind the pointer. With the indicator in normal operating position the pointer shall move in such a manner that it will remain horizontal at any center at any point of deflection up or down from center, and shall deflect up or down in agreement with the polarity of direct current received from the flight direction computer. The dimensioned portion of the horizontal pointer shown on Figure 4 shall be color No. 33695 of FED-STD-595. NOTE: The flight director computer referred to herein is designed so that current to the pointer movement is limited to 2.5 ma maximum when in a mode when the pointers are being used. The application of 13.5 ma as a step unit input shall drive the pointer out of view and shall not damage the meter movement. Both pointer movements shall be rated as continuous duty for this condition.

*3.5.7.2.1 Deflection - The maximum deflection upward shall be physically limited to 5/8 inches. The design shall be such that the pointer can be deflected down until it disappears from view behind a mask. The direction of deflection for a given input signal shall be as specified in Table III.

*3.5.7.2.2 Response - The horizontal pointer response shall be linear in degree with respect to current within 7.5 percent of the proportionate full scale value. A deflection of 1/2 inch, measured from the center of the dial to the pointer along a vertical line, shall require 2.2 ma ± 7.5 percent; 2.5 ma shall not deflect the pointer more than 5/8 inch. Application of 10 ma of the proper polarity shall drive the pointer from view behind its mask at the bottom of the display. The application of 13.5 ma as a step input to drive the pointer from view shall not damage the meter movement. (Continuous duty capability required.)

3.5.7.2.3 Resistance - Resistance across the horizontal pointer input terminals shall be 1,000 ohms ± 3 percent at 25°C.

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3.5.7.2.4 Damping - The pointer mechanism shall be damped so that there shall not be more than 1-1/2 percent overshoot.

*3.5.7.2.5 Response Time - Response time (see 6.5.1) of the horizontal pointer shall be maximum of .45 second.

3.5.7.3 Pointer Deflection Test - A self test button shall be provided on the face of the indicator. Depressing the test button shall cause the vertical and horizontal pointers to change from the stowed position to indicate zero deflection, with respect to the aircraft symbol. Depressing the test button shall also result in a rate of turn indication of two full needle widths.

*3.5.8 Pitch and Roll Outputs - The pitch and roll outputs shall give standard synchro type output with a no-load of 11.8 ± .4 volts line-to-line and a no-load phase shift which shall not exceed 8°. The roll output shall be continuous through 360°. The pitch output shall be continuous to 78° dive and 92° climb. Each synchro shall follow the respective gimbal movement to a one-to-one ratio.

3.5.8.1 Synchro Transmitter Characteristics - Each transmitter shall be capable of operating up to four Bendix, Type AY500-5 or equivalent, control transformers. When loaded with four loads the electrical characteristics of each transmitter shall not vary by more than the following:

(1) The voltage drop of the nominal unloaded line-to-line output shall not exceed 0.6 volt with the rotor leads of each control transformer open.

(2) The angle measured between the exciting voltage of each transmitter rotor and the voltage in any of the three stator leads of each of the four control transformers shall not exceed 3.5°. This measurement shall be made with the rotor leads of the four control transformers open.

(3) Nulls measured from one up to four loads shall not exceed 0.06 volts. The conditions specified in (1) and (2) shall apply to the pitch and roll transmitters at room temperature.

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*3.5.8.2 Synchro Zeroing and Sensing - The synchros shall be zeroed and their lead terminals shown in Table III and tested in accordance with paragraph 4.8.5.4.

*3.5.8.2.1 The synchro transmitter leads are defined with reference to the external electrical receptacle pins for the above procedure in Table III.

3.5.9 Indicator Case - The outline dimensions, of the attitude indicator case shall conform to Figure 8, sealing compound conforming with MIL-S-8802 shall be used where applicable. The case shall meet the requirements of the environmental tests.

3.5.9.1 Cover Glass - The cover glass shall be flat and free from discolorations, scratches, and striae which interfere with normal reading of the indicator and shall comply with MIL-G-81704 Type I and DD-G-451 Type II, Quality AA.

3.5.9.2 Anti-Reflection Coating - All reflecting glass surfaces of the cover glass and lighting wedge shall be covered with a reflection reduced coating that meets the requirements of MIL-C-14806 in addition to withstanding the environmental requirements specified herein. The humidity requirements shall be governed by paragraph 4.8.34.

3.5.10 Lighting - The attitude indicator shall be provided with integral lighting in accordance with MIL-L-25467. The maximum current required for the lighting circuit in the indicator shall not exceed 1 ampere AC or DC. The lighting shall be subject to approval of the procuring activity.

3.5.10.1 Prime Standard - The internal lighting shall be comparable to and shall balance in brightness with a prime standard indicator. The prime standard shall be either an attitude director conforming to the requirments of this specification or a lighted model. The prime standard shall be equipped with red lighting and shall be used as the reference for acceptance of all production indicators.

3.5.10.2 Visual Comparison - The general overall lighting level of the indicator shall fall within the limits of the prime standard when energized at the voltage levels specified.

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*3.5.11 Power Consumption - The power consumption of the indicator shall not exceed 19 VA when operating from the emergency bus as a vertical gyro reference. Power consumption shall not exceed 19 VA when operating with interfacing equipment from the main bus. The power factor shall be maintained between .85 and 1.00.

3.5.12 Electrical Power Variation - The indicator shall operate properly in accordance with the requirements of MIL-STD-704.

3.5.13 Power Failure Indication - A power failure warning flag shall be incorporated within the indicator and shall be tested according to 4.8.10. The warning flag shall come into view within 1 second as shown in Figure 1, in the event of loss of power to all or any phase of the gyroscope motor. The warning flag shall be fluorescent red color 38905 of FED-STD-595, or equal, and the lettering "OFF" shall be black color 37038 of FED-STD-595. The fluorescent red flag shall be illuminated under night lighting conditions and the lighting shall not be entirely dependent upon one light source. The flag mechanism shall be rugged and designed to withstand hang-up under normal service use.

3.5.14 Electrical Connector - The electrical connector shall conform to MIL-C-81511 and shall be located in the lower corner on the rear of the instrument. The indicator connector shall mate with an M81511/Q6E14-37S1 connector. The physical position of the polarizing key on the connector, with respect to the indicator case, shall be as shown in Figure 8. The electrical connections and the connector wiring shall be in accordance with Table V of this specification.

*3.5.15 Ball Bearings and Lubrication - Bearing and lubrication shall be in accordance with MIL-B-81793, unless otherwise specified.

*3.5.15.1 Rotor Bearing Lubrication - The rotor bearings shall be lubricated with grease MIL-G-81322, MIL-G-81937 or Minipure with particle size not exceeding 50 microns for two percent and 5 microns for 98 percent volume. The bearing separators shall be impregnated with Mobil oil 743A or equivalent.

*3.5.15.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall be in accordance with MIL-L-81846 or Mobil oil 743A.

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*3.5.16 Barrier Film Bearing Coating - All bearings, except spin motor bearings, shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings, including chamfered races or corners, and including interfacing parts where applicable, shall be barrier coated.

*3.5.17 Ball Bearings - Ball bearings shall be ABEC-5 grade or better, teflon separators shall be used on gimbal bearings, where the bearing design permits. Spin rotor bearings shall have porous separators impregnated with lubricant MIL-L-81846 and shall be ABEC-7 or better.

3.5.17.1 Ball Bearing Pre-Loading - Means shall be provided for pre-loading and alignment without causing eccentric loading unless otherwise authorized.

3.6 Controlled Environment - The requirments of FED-STD-209 for clean room Class 10,000 and work station Class 100 environment shall be complied with.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) (See 4.3)

(2) Acceptance Tests (See 4.4)

(3) Life Tests (See 4.5)

*4.3 Preproduction (First Article Tests) - Preproduction Tests shall be made on nine instruments representative of the production instruments to be supplied under the contract. Preproduction Tests shall be accomplished

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using the test procedures of 4.6. The government inspector and the procuring agency shall be advised when the tests are to be conducted so that a representative may be designated to witness the inspection when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the procuring activity.

*4.3.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

*4.3.2 Scope of Preproduction (First Article) Tests - Preproduction tests shall include all tests listed in 4.8, and the reliability test for reliability qualification and such other tests deemed necessary by the procuring activity to determine that the instrument meets all requirements of this specification and the contract.

*4.3.2.1 Test Samples - Three samples shall be subjected to all the tests of 4.8 and the other six shall be subjected to the qualification reliabiltiy test of paragraph 4.4.3.1. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

*4.3.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

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*4.3.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

*4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing all tests except Sampling Plan "B" Tests which may be conducted at a Government laboratory designated by the procuring activity. Testing shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing detailed quantitive results for all tests required by this specification, signed by an authorized representative of the laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

(1) Individual Tests
(2) Sampling Plan "A" Tests
(3) Sampling Plan "B" Tests
(4) Reliability Assurance Production Acceptance Phase
(5) Special Tests

*4.4.1 Individual Tests - Each indicator submitted for acceptance shall be subjected to and pass the following individual tests:

(1) Examination of Product	4.8.1
(2) Initial Erection	4.8.2
(3) Final Erection	4.8.3
(4) Spin Axis Verticality Test	4.8.4
(5) Synchro Pick-off Requirement	4.8.5
(6) Erection Rates	4.8.6
(7) Erection Cut-off Angle	4.8.7
(8) Gimbal Freedom	4.8.8

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(9) Caging Knob Operation	4.8.9	
(10) Power Warning Flag Test	4.8.10	
(11) Pitch Trim Test	4.8.12	
(12) Power Consumption	4.8.13	
(13) Rotor Coast Time	4.8.14	
(14) Inclinometer Tests	4.8.15	
(15) Rate-of-Turn Tests	4.8.16	(except 4.8.16.1 and 4.8.16.3)
(16) Vertical & Horizontal Pointer Tests	4.8.17	(except 4.8.17.2 and 4.8.17.4)
(17) Visibility of Display	4.8.18	
(18) Drift Rate (where applicable)	4.8.20	
(19) Leak Test (where applicable)	4.8.21	
(20) Insulation Resistance	4.8.22	

*4.4.2 Sampling Plans - The sampling plans shall consist of Sampling Plan "A" and Sampling Plan "B" Tests. The test samples which have been subjected to Sampling Plan "A" Tests shall not be delivered on contract until they have complied with the requirements of paragraph 4.5.4. Sampling Plan "B" Test units shall be retained by the procuring activity.

*4.4.2.1 Sampling Plan "A" Selection - Samples shall be selected at random in accordance with the following schedule:

Quantity of Equipments for Acceptance	Quantity to be Selected for Inspection
First 10	1 (Zero when Sampling Plan "B" is invoked)
Next 50	1
Next 75	1
Next 100	1
Each additional 100 or fraction thereof	1

When a defective instrument is detected, no item from those still on hand or later produced shall be accepted until the extent and cause of failure has been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of each 15 from those still on hand, and proceed with sampling as indicated above when the next quantity is offered for acceptance.

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*4.4.2.1.1 Sampling Plan "A" Tests - Each sample selected for Sampling Plan "A" Tests shall be subjected to the following tests starting immediately upon selection:

(1) Individual Tests	4.4.1
(2) Rate-of-Turn Response Time	4.8.16.1
(3) Rate-of-Turn Indicator Dynamic Balance	4.8.16.3
(4) Vertical and Horizontal Pointers Response Time	4.8.17.2
(5) Dynamic Balance, Vertical and Horizontal Pointers	4.8.17.4
(6) Fogging	4.8.19
(7) Power and Frequency Variation	4.8.23
(8) Magnetic Effect	4.8.24
(9) Vibration Error	4.8.25
(10) Low Temperature Operation	4.8.26
(11) High Temperature Operation	4.8.27

*4.4.2.2 Sampling Plan "B" Instructions - One sample shall be selected at random from the first 10 produced on contract and submitted within 10 days after manufacture. The sample shall be forwarded at the contractor's expense to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

Indicator Gyro-
Submitted by (Manufacturer's Name, Date) for Production Acceptance Sampling Plan "B" Inspection of MIL-I-81683A(AS) with Contract No._____ Manufacturer's Part No. _____

4.4.2.2.1 Sampling Plan "B" Approval - Approval of Sampling Plan "B" instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

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*4.4.2.2.2 Sampling Plan "B" Tests - Each sample selected for Sampling Plan "B" testing shall be subjected to the following tests:

(1)	Sampling Plan "A" Tests	4.4.2.1.1
(2)	High Temperature Exposure	4.8.28
(3)	High Altitude-Low Temperature	4.8.29
(4)	Loop Error	4.8.30
(5)	Acceleration	4.8.31
(6)	Impact Shock	4.8.32
(7)	Vibration Failure	4.8.33
(8)	Humidity	4.8.34
(9)	Salt Spray	4.8.35
(10)	Fungus	4.8.36
(11)	Sand and Dust	4.8.37
(12)	Radio Noise Interference	4.8.38
(13)	Temperature Shock	4.8.39
(14)	Performance	4.8.40
(15)	Explosive Test	4.8.41
(16)	Internal Examination	4.8.42

*4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted in accordance with paragraphs 4.4.3.1 and 4.4.3.2.

*4.4.3.1 Preproduction (First Article) Reliability Qualification - Prior to acceptance of equipment under contract or order, equipment shall be tested as outlined in MIL-STD-781 under section entitled "First Article (Preproduction) Reliability Qualification". The following conditions apply:

Equipment Class - Class II, MIL-E-5400; Category 4, MIL-STD-781

Upper Test MTBF $\emptyset_0$ - 2000 hours

Lower Test MTBF $\emptyset_1$ - 1000 hours

Test Plan - Fix length test plan 20-6, Appendix C, page 115, Figure C-19 of MIL-STD-781

Environmental temperature extreme shall be +55°C and -44°C. The vibration level shall be .5g.

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*4.4.3.1.1 Individual Tests and Increase Tolerance - After each 150 hour cycle of operation the indicators shall be subjected to and meet the requirements of the following Individual Tests:

(1) Examination of Product	4.8.1
(2) Initial Erection	4.8.2
(3) Final Erection	4.8.3
(4) Synchro Accuracy	4.8.5.1
(5) Synchro Outputs	4.8.5.3
(6) Erection Rates	4.8.6
(7) Power Warning Flag Test	4.8.10
(8) Inclinometer Test	4.8.15
(9) Rate of Turn	4.8.16.4
(10) Vertical and Horizontal Pointers	4.8.17.3
(11) Illumination of Display	4.8.18

Increase Tolerance

	500 - 1000 Hrs.	1000 - 1500 Hrs.	1500 Hrs.
4.8.3	± .65°	± .75°	± .85°
4.8.5.1	± .65°	± .75°	± .85°
4.8.6	T.9 ± .75°/min.	T.9 ± .9°/min.	T.9 ± 1.05°/min
4.8.10	-----	1.5 sec.	3 sec.
4.8.16.4	± .020 inch	± .023 inch	± .024 inch
4.8.17.3	± .020 inch	± .023 inch	± .026 inch
4.8.18	Prime std comparison 2.8 to 2.2 volts	Lamps shall be lighted	Lamps shall be lighted

*4.4.3.2 Reliability Assurance Production Acceptance Phase - Reliability Assurance Production Acceptance testing shall be conducted through burn-in. Each equipment produced shall pass a minimum 150 hours Burn-In Test. Any failure occuring during the last 50 hours which requires component changes or major recalibration, shall be subjected to 50 hours after completion of repairs. Units accumulating 400 hours or more shall be overhauled (spin bearing replaced and other wear parts replaced as necessary) and resubmitted to the minimum 150 hour Burn-In Test. During the burn-in period, the units shall be operated and subjected to the environmental conditions of +55°C and -44°C, and a vibration level of 0.5g. At the conclusion of each month the test

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results shall be forwarded to the procuring activity upon request. In the event the test data shows a reduction in test failures, the procuring activity reserves the right to reduce the burn-in time.

NOTE: Each equipment shall be subjected to and meet the requirements of the Individual Tests prior to and after the Burn-In Test.

*4.4.3.3 Test Details - The test details, such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristic to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures of 4.6 to be submitted to and approved by the procuring activity prior to the beginning of the acceptance test phase of the reliability assurance tests.

*4.4.3.4 Prior Approval - If production approval is granted by the procuring activity prior to completion of the Reliability Qualification Tests, then the contractor shall be responsible for correction of any and all defects discovered during the Reliability Qualification testing. The contractor is further required to develop and retrofit corrective action on all units in work or delivered. If at any time during the Reliability Qualification testing a reject decision is reached, delivering of equipment shall be stopped and the procurement activity notified.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of equipment, and to assure adequate quality control. The equipment to be subjected to special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipment for special tests shall be made as follows:

(1) On an early indicator after any (material or engineering) change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the Procuring Activity.)

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4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the qualification tests shall be used where applicable. Where such tests are not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include the test in an individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

*4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing this test. The test shall be for 1500 hours duration and shall be conducted on equipments that have passed the Reliability Assurance Production Acceptance Phase. The test shall be conducted under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
Each additional 500 or fraction thereof	1

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4.5.1 Conditions for the Life Test - The life test shall be conducted under the standard conditions of 3.3.8, and the simulated service conditions as follows:

Conditions	Test Condition	Time (approx. hrs.)
1	3° (±1.5°) Total Angle, Pitch, Roll & Yaw Motion	450
2	15° (±7.5°) Total Angle, Pitch, Roll & Yaw Motion	450
3	Vibration - 0.002 to 0.005 inch double amplitude and a frequency of 25 to 33 Hz.	450
4	Tests conducted in accordance with this specification as specified by the procuring activity.	150

*4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. A daily visual observation shall be made for proper operation.

*4.5.3 Performance Check - At approximately 100-hour intervals, the indicator shall be subjected to and meet the requirements of the individual tests:

(1) Initial Erection (4.8.2)
(2) Final Erection (4.8.3)
(3) Synchro Pick-off Requirements (4.8.5)
(4) Erection Rates (4.8.6)
(5) (Rate-of-Turn) Sensitivity (4.8.16.4)
(6) Power Consumption Test (4.8.13)
(7) (Vertical and Horizontal Pointers) Response (4.8.17.1)

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4.5.3.1 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part Type Number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failures.

4.5.3.1.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred.

*4.5.4 Reconditioning of Tested Equipment - Equipment which has been subjected to reliability qualification, acceptance, and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

*4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary.

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4.7 Test Conditions -

4.7.1 Atmospheric Conditions - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of 3.3.8.

4.7.2 Attitude - Except as otherwise specified below, all tests shall be performed under the following conditions:

(a) The indicator shall be mounted and mechanically secured to a fixture so that the longitudinal axis of the indicator is 7° connector end down. Wherever the term "indicator" is used in the following paragraphs, it shall be understood to refer to the indicator mounted on the fixture as described above.

(b) All visual tests of indicator performance shall be observed from a distance of 2 feet in front of the glass cover, along the center line of the indicator. The pitch scale graduations from 20° to -20°, the bank index, the numerals and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 25° with a perpendicular to the cover glass and whose small diameter is the aperture of the instrument case.

4.7.3 Input Power - The input power shall be in accordance with 3.3.8.

4.7.4 Test Equipment and Connections - Special test equipment and connections thereto shall be as specified. Whenever practical, the test equipment used in performing the tests shall not affect the performance of the equipment under test.

4.7.5 Test Warm-Up Time - For all tests except initial and final erection tests, a warm-up period of 5 minutes under standard conditions and 10 minutes under extreme service conditions shall be allowed before commencing any test.

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4.8 Test Methods -

4.8.1 Examination of Product - Each indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, markings, and dimensions.

4.8.2 Initial Erection - The indicator shall be mounted on a surface level within 1 minute of arc in roll and plus 4° ± 1 minute of arc in pitch. Apply power for a minimum of 50 seconds. Between 50 and 60 seconds, the caging knob shall be pulled until the horizon line is stabilized. After returning the knob quickly, the indicator shall indicate 0° ± 1° in roll and 4° ± 1° in climb.

*4.8.3 Final Erection - The indicator shall be mounted level in roll and 4° climb on a pitch, roll, and yaw mechanism adjusted to cause a 1.5° displacement of the indicator in pitch, roll, and yaw, at a frequency of 5 to 7 oscillations per minute. After two minutes of operation on this mechanism the indicator shall be immediately removed and mounted on a surface level within 1 minute of arc. With the pitch trim knob set at the zero indication, the indicator shall indicate vertical within 0.5° in both pitch and roll.

4.8.4 Spin Axis Verticality Tests -

*4.8.4.1 Rotation in Azimuth - Immediately after the final erection period, the indicator shall be rotated 180° in azimuth, the change in pitch and roll shall not indicate more than 1°.

4.8.5 Synchro Pick-off Requirements -

4.8.5.1 Accuracy - With the indicator mounted on a level surface and operating normally the pitch and roll outputs shall not exceed 0.5°.

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4.8.5.2 Null - The open circuit null voltages of the pitch and roll outputs shall not exceed .060 volt rms across pins 16 to 17 and 13 to 14, respectively.

4.8.5.3 Maximum Output - The maximum open circuit pitch and roll outputs shall be 11.8 ± .4 volts line-to-line and the phase shift shall not exceed 8°.

4.8.5.4 Zeroing and Sensing - The synchros shall be zeroed and their lead terminals shown in Table V connected for the following procedure:

(1) Connect terminals 15 and 18 to pin 10 with the gyroscope gimbals at zero displacement about their axes. The voltage between pins 13 and 14 and between pins 16 and 17 shall be at null.

(2) With 13 and 14 strapped, the voltage between this point and 1 shall be less than 115 volts.

(3) With 16 and 17 strapped, the voltage between this point and 1 shall be less than 115 volts.

(4) With 13 and 14 unstrapped and the gyro case displaced in right bank about the roll axis with respect to the roll gimbal, the voltage between 13 and pin 10 shall increase before decreasing.

(5) With 16 and 17 unstrapped and the gyro case displaced in climb about the pitch axis with respect to the pitch gimbal, the voltage between 16 and pin 10 shall increase before decreasing.

*4.8.6 Erection Rates - The gyro shall be capable of erecting in pitch and roll at a rate of 1.3 to 2.5° per minute. Roll erection shall be cut off when the spin axis displacement exceeds 5 ± 3/4° in roll. The indicator shall be mounted on a pitch, roll, and yaw mechanism adjusted for 1.5° displacement and at a frequency of 5 to 7 oscillations per minute for the following tests:

*4.8.6.1 Low Angle Erection

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*4.8.6.1.1 Low Angle Erection - Roll Axis - With the indicator oriented towards the geographical West, the gyro spin axis shall be inclined 4° toward the geographical North. The time required for the gyro to erect to within 2° of vertical shall be between 48 seconds and 1 minute 32 seconds. With the spin axis inclined 4° to the South, the same limits shall apply. The pitch indication shall not change more than 1° while the roll axis is erecting.

*4.8.6.1.2 Low Angle Erection - Pitch Axis - With the indicator oriented towards the geographical North, the gyro spin axis shall be inclined 6° towards the North. The time required for the gyro to erect to within 2° of vertical shall be between 1 minute 36 seconds and 3 minutes 4 seconds. With the spin axis inclined 6° towards the South, the same limits shall apply. The roll indication shall not change more than 1° while the pitch axis is erecting.

*4.8.6.2.1 High Angle Erection - Roll Axis - With the indicator oriented towards the geographical West, the gyro spin axis shall be inclined 30° towards the North. The roll indication shall not change more than 3-1/2° in a 4 minute period. The test shall be repeated with the gyro spin axis inclined 30° towards the South. The same results shall be obtained.

*4.8.6.2.2 High Angle Erection - Vertical - With the roll axis of the indicator oriented along the vertical and the spin axis oriented towards the geographical North, the gyro shall be aligned to indicate 0° in pitch and roll. The roll indication shall not change more than 2° in a 32 second period.

*4.8.7 Roll Axis Cut-off Angle - With the indicator oriented towards the geographical West, the gyro spin axis shall be inclined towards the North. The roll angle that removes roll erection shall be determined. The limits shall be 5 ± 3/4 degrees from vertical. Repeat the test with the gyro spin axis inclined towards the South. The same limits shall apply.

*4.8.8 Gimbal Freedom Test - With the indicator operating normally, displace the gyro 360° in roll, 89° in climb, and 75° dive. The spheroid and roll index shall follow smoothly without noticeable sticking or binding in either axis. The test shall be repeated with power removed.

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4.8.9 Caging Knob Operation - The caging knob shall be pulled and held in the fully extended position until the horizon line is stabilized. The knob shall then be rotated to the extreme clockwise position; the indicator shall remain caged and the knob shall remain locked in the extended position until it is released. No malfunction shall result from sudden release of knob.

4.8.10 Power Warning Flag Test -

4.8.10.1 Power Interruption - After a minimum of 5 minutes of operation the power shall be removed. The power warning flag shall appear when power is removed and shall disappear when power is restored, within the period specified in 3.5.13. The same test shall be repeated, removing the power on each phase individually.

4.8.10.2 Low Voltage - When the 115 volt, 400 cps power is decreased to 103 volts, the flag shall not appear.

4.8.12 Pitch Trim Tests -

4.8.12.1 Pitch Trim Adjustment - When the knob is rotated to the stop in the clockwise direction, the miniature airplane shall move upward a minimum of 5°. When the knob is rotated to the stop in the counter-clockwise direction, the miniature airplane shall move downward to indicate a minimum of 10° dive.

4.8.13 Power Consumption Test - The power consumption of the indicator shall be measured with suitable meters. The AC consumption shall be in accordance with paragraph 3.5.11. Starting power for the indicator shall not exceed 40 VA.

*4.8.14 Rotor Coast Time - With the indicator operating a minimum of 10 minutes, the power shall be removed. The pitch and roll indication shall not change more than 6° in 3 minutes.

4.8.15 Inclinometer Tests -

*4.8.15.1 Inclinometer Sensitivity - With the dial inclined 7 degrees, the indicator shall be rotated about the roll axis to the right until the ball is just short of its limit. The angle of rotation shall be 10° ± 2°. The test shall be repeated rotating the instrument to the left. The same tolerance shall apply.

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4.8.15.2 Inclinometer Zero Position - The indicator shall be positioned with the upper edge of the mounting bezel horizontal, and the indicator shall be gently tapped. The ball shall rest at the zero mark of the slip indicator within 1/32 inch.

*4.8.15.3 Inclinometer Friction - The indicator shall be tipped at a rate of 4 ± .2°/second to either side of the vertical plane of the dial. The ball shall roll smoothly when the indicator is gently tapped.

4.8.16 Rate-of-Turn -

*4.8.16.1 Rate-of-Turn Response Time - The response time of the rate-of-turn pointer shall be measured and shall be in accordance with 3.5.6.5. For this test, pins 26 and 27 shall be shunted externally with a resistance of 1000 ohms to allow for the shunting effect of another indicator in parallel with the usual source. At least 5 seconds shall be allowed in order to establish the calibration of the 100 percent mark. The voltage source shall be steady and applied to the above circuit through at least 1,000 ohms.

*4.8.16.2 Rate-of-Turn Indicator Static Balance - The indicator shall be placed in a stationary position and the pointer checked with no power applied. The deviation of the pointer from initial position shall not exceed 1/32 inch for any stationary position.

*4.8.16.3 Rate-of-Turn Indicator Dynamic Balance - The indicator shall be accelerated along its lateral axis with a force of 4g. The rate-of-turn pointer shall not vary by more than .100 inch from the initial position.

*4.8.16.4 Rate-of-Turn Indicator Sensitivity - The indicator shall be placed in the normal, stationary position. The instrument shall indicate the deflections for the given currents in Table IV. The deflection of the pointer shall conform to the tolerances in Table IV for clockwise and counterclockwise rates. The movement of the pointer shall be smooth throughout its range. The over travel in both directions shall be a minimum of 1/2 the pointer width.

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TABLE IV. Pointer Deflection.

Rate-of-Turning Degrees Per Minute	Deflection of Pointer In Inches	Input Current Microamperes
0	0 ± .017	0
90	.156 ± .029	500
180	.312 ± .041	1000

4.8.17 Vertical and Horizontal Pointers -

4.8.17.1 Response - The response of the pointers shall be measured and shall be in accordance with 3.5.7.1.5 and 3.5.7.2.2. A step input equivalent required to drive each pointer out of view shall be applied ten consecutive times. No damage or sticking shall result.

*4.8.17.2 Response Time - The response time of the pointers shall be measured and shall be in accordance with 3.5.7.1.2 and 3.5.7.2.2. Each meter movement shall be connected in parallel with a like meter movement of a 1000 ohm simulating the shunting effect of another indicator in parallel. The signal source impedance shall be 1000 ohms minimum.

*4.8.17.3 Balance - With the indicator in a normal upright position the pointers shall bisect the center dot of the miniature aircraft within ± .017 inch. The pointers shall not be deflected by more than 1/32 inch as the indicator is rotated 360 degrees in roll and pitch.

*4.8.17.4 Dynamic Balance - The indicator shall be accelerated for one minute with a force of 4g along the respective axes in the following positions:

(a) With the indicator in its normal attitude facing the center of rotation, accelerate along its longitudinal axis.
(b) With the top of the indicator facing the center of rotation, accelerate along its vertical axis.
(c) With the left side facing the center of rotation, accelerate the indicator along the lateral axis.

The pointers shall not vary by more than .100 inches from the initial position.

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4.8.17.5 Self Test Pointers - With the indicator operating normally and the vertical and horizontal pointers stowed, depressing the test button shall cause the horizontal pointers indicate zero and the turn rate pointer to deflect .312 ± .050 inches to the right. The test button should be labeled as in Figure 1.

*4.8.18 Illumination of Display - The indicator shall be tested for comparison with the prime standard in accordance with Specification MIL-L-25467.

4.8.18.1 Prime Standard - Using parts which are identical to the parts to be used in production indicators, an indicator shall be constructed which meets all display requirements of Specification MIL-L-25467. The indicator model and recorded lighting data shall be submitted to the procuring activity for approval; and if approved, the indicator model shall be certified as a prime standard.

4.8.18.2 Prime Standard Life - A minimum of 12 average areas covering all representative areas of the face of the prime standard shall be measured for brightness and color prior to submission, at least every 50 hours of operation thereafter. A change of +/-20% brightness or +/-5% color change in any area using the initial data as a reference voids the prime standard certification.

*4.8.18.3 Production (Visual Comparison) - With 2.50 ± .01 volts simultaneously applied to the indicator under test, and to the prime standard, like areas of the warning flag, moving background, numerals, indicia, markings, miniature airplane, etc., shall be compared. Overall uniformity, similarity, and brightness of the test indicator to that of the prime standard shall be observed. Like comparison of the test indicator to the prime standard shall allow for acceptance.

If the brightness is dissimilar in any individual area, 2.40 and 2.60 volts shall be applied consecutively to the prime standard.

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The brightness of the production indicator at 2.50 ± .01 volts shall fall within the brightness of the prime standard when set at 2.40 and 2.60 volts.

A color comparison test shall be made with the prime standard and production indicator energized with 5.0 ± .01 volts. The indicators shall be identical in color.

If any questionable areas still exist, they shall be measured with a photometer in accordance with the requirements of MIL-L-25467. If the production indicator fails this test, it shall be rejected and returned for rework.

*4.8.19 Fogging - The indicator shall be energized and in a +71° ± 2°C controlled ambient for a minimum period of 1 hour. After this period and while at this temperature, an ice cube shall be rubbed on the indicator glass face for a period of 1 minute ± 5 seconds. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Fogging shall not be present on the inner surface of the cover glass to the extent that visibility of the indicator shall be the cause of rejection. The ice cube (one inch cube minimum) shall be at a temperature of approximately 0°C to -10°C.

4.8.20 Drift - Where applicable, the free drift rate of the displacement gyro shall not exceed 15 degrees per hour after compensation for earth's rate with the spin axis vertical or tilted off the vertical 20 ± 1/2 degrees.

4.8.21 Leakage - Where applicable, the indicator shall be tested for leaks by means of a mass spectrometer type helium detector. The gas or mixture of gases in the case adjusted to a pressure differential of one atmosphere, shall not have a leakage rate that would permit more than 10 percent contamination of the filling medium after 1000 hours.

*4.8.22 Insulation Resistance - With a potential of 500 volts direct current applied for 5 seconds between isolated circuits and between circuits and the case, the resistance shall be greater than 50 megohms.

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*4.8.23 Power and Frequency Variation - The input voltage and frequency shall be varied within the following limits and combinations.

	A.C. Volts	Frequency (Hz)
(a)	103 to 127	420
(b)	103 to 127	380
(c)	103	380 to 420
(d)	127	380 to 420

The indicator shall be subjected to and meet the requirements of individual tests 4.8.2, 4.8.3, 4.8.4, 4.8.5.1, 4.8.5.2, and 4.8.5.3 (except that the output voltage shall not change more than 13%).

*4.8.23.1 Power Transients - The indicator shall be properly connected with AC power applied and subjected to the transient power conditions of MIL-STD-704. This test shall not damage the indicator. The indicator shall then be subjected to and meet the requirements of the Individual Tests.

4.8.24 Magnetic Effect - The indicator shall be properly connected and power applied. The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 12 inches from the bar magnet. The compass shall have its compensating magnets removed and shall be set-up in a uniform magnetic field whose horizontal intensity is between 0.17 and 0.19 oersted. The indicator shall be revolved in a horizontal plane which is perpendicular to the axis of the bar magnet. The indicator shall be held in positions 0, 45, 90, 135, 225, 270, and 315°. At each of these positions, the indicator shall be rotated 360° about its horizontal axis. The deflection of the compass at any of the specified positions shall not exceed 5°. This test shall be repeated with no power applied.

*4.8.25 Vibration Error - The indicator, properly connected, shall be vibrated as specified in paragraph 4.2 of Specification MIL-T-5422 to the requirements of

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Curve II Figure V except one sweep of 5 to 50 to 5 Hertz shall be completed in each axis. During the vibration period, the indicator shall meet the requirements of 4.8.25.1, 4.8.25.2, 4.8.25.4 through 4.8.25.8.

4.8.25.1 Indicator Sphere Roll and Pitch Oscillation - The maximum total amplitude of the indicator sphere roll and pitch oscillation throughout the frequency range shall be noted and shall not exceed 1° in double amplitude in roll and pitch. Anti-backlash gears shall be provided, where required, to eliminate oscillation and jitter in the indicator presentation.

*4.8.25.2 Indicator Sphere Roll and Pitch Variation - The maximum variation in roll and pitch during vibration, from the original indication before vibration, throughout the frequency range, specified in 4.8.25, shall be noted and shall not exceed ± 2° in roll and pitch.

4.8.25.3 Oscillation and Variation Tests - May be conducted at any normal flight attitudes.

4.8.25.4 Inclinometer Accuracy - The inclinometer reading shall not differ at any time by more than ± 1/16 inch from the reading obtained when the indicator is at rest.

4.8.25.5 Rate-of-Turn Variation - Throughout the frequency range, the maximum variation of the rate-of-turn pointer shall not exceed 1/32 of an inch maximum.

*4.8.25.6 Pointer Oscillation - The maximum total amplitude of the vertical and horizontal pointers throughout the frequency range shall not exceed 1/2 the width of the respective pointer at the center and two times at the ends of the pointers.

4.8.25.7 Pointer Variation - Throughout the frequency range the maximum variation of the vertical and horizontal pointers shall not exceed 1/2 the width of the respective pointer.

4.8.25.8 Pitch and Roll Pick-Offs - There shall be no indication of output signal discontinuities throughout the frequency range.

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*4.8.26 Low Temperature Operation - The indicator, properly connected with no power applied, shall be tested in accordance with Step 2 of the Temperature-Altitude Test, of MIL-T-5422, except as modified below. At the end of 4 hours and while at reduced temperature, power shall be applied for 10 minutes. At the end of the 10 minute period, but still at reduced temperature, the indicator shall be subjected to and shall meet the requirements of 1/4.8.2, 1/4.8.3, 4.8.5.2, 4.8.5.3, and 4.8.6 (except final erection shall be within ± .65°, synchro null .070 volts maximum, and erection time within 32 seconds and 2 minutes for roll and between 1 minute 4 seconds and 4 minutes for pitch). The rate-of-turn pointer, vertical pointer, and horizontal pointer shall be checked only for operation without evidence of sticking or binding and that the null indication has not shifted more than ± 1/64 inch from initial position. There shall be no slip indicator leakage as a result of this test. The indicator shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours after which it shall be operated and shall meet the requirements of the individual tests. There shall be no damage which would adversely affect subsequent operation.

*4.8.27 High Temperature Operation - The indicator, properly connected, shall be tested in accordance with Step 6 of the Temperature-Altitude Test, Class II of MIL-T-5422. At the end of the 4 hour period, and while still at the elevated temperature, the indicator shall be subjected to and shall meet the requirements of the following: 1/4.8.2, 1/4.8.3, 4.8.5.2, 4.8.5.3, and 4.8.6 (except final erection shall be within ± .65°, synchro null .070 volts maximum, and erection time within 32 seconds and 2 minutes for roll and between 1 minute 4 seconds and 4 minutes for pitch). The rate-of-turn pointer, vertical pointer, and horizontal pointer shall be checked only for operation without evidence of sticking or binding and that the null indication has not shifted more than ± 1/64 inch from initial position. There shall be no slip indicator leakage. The indicator shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours, after which, it shall be subjected to and shall meet the requirements of the individual tests.

Note 1/ See paragraph 4.7.5

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*4.8.28 High Temperature Exposure - The indicator shall be tested per MIL-T-5422, temperature and altitude tests of Step 6, except as modified, for a period of 24 hours. The system shall be properly connected, but no power shall be applied during the test. During the last 15 minutes of the 24 hour period, normal power shall be applied. At the end of the 24 hour period and while still at the elevated temperature, the indicator shall be subjected to and shall meet the requirements of 4.8.2, 4.8.3, 4.8.5.2, 4.8.5.3, and 4.8.4 (except final erection shall be within ± .65°, synchro null 0.070 volts maximum, and erection time within 32 seconds and 2 minutes for roll and between 1 minute, 4 seconds and 4 minutes for pitch). The rate-of-turn pointer, vertical pointer, and horizontal pointer shall be checked only for operation without evidence of sticking or binding and that the null indication has not shifted more than ± 1/64 inch from initial position. There shall be no slip indicator leakage. The indicator shall then be returned to standard conditions and allowed to stabilize for a period of 4 hours, after which, it shall be subjected to and shall meet the requirements of the individual tests.

*4.8.29 High Altitude-Low Temperature - The indicator, properly connected but with no power applied, shall be subjected to Step 3. Temperature and Altitude Tests, of MIL-T-5422, except as modified for a period of 24 hours. At the end of the exposure period, power shall be applied for a maximum of 10 minutes to operate the indicator. After caging, it shall be observed for 30 minutes and shall indicate the vertical within ± 2°. After the equipment has returned to room temperature and pressure it shall be subjected to and shall meet the requirements of the individual tests. 1/

*4.8.30 Loop Error - The indicator shall be mounted on a fixture permitting rotation about the pitch axis. The indicator shall be rotated one complete revolution (loop) at a rate of 12 ± 1° per second and returned to the normal position. The error in either pitch or roll shall not exceed 3°. Furthermore, rotating the indicator in either direction at a rate of 90 ± 5° per second shall not cause damage to the indicator. The error induced by roll maneuvers at a rate of 300° ± 10° per second shall not exceed 1°. Furthermore, rotating the indicator in either direction at the above rates shall not cause damage to the indicator.

Note 1/ See paragraph 4.7.5

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4.8.31 Accelerations - The indicator not operating shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of the 2 axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20g in each of the above axes for a period of 1 minute. At the end of this acceleration, the indicator shall meet the requirements of the Individual Tests. No damage to the indicator shall result from this test.

4.8.32 Impact Shock - The indicator shall be tested in accordance with the Shock Test of Specification MIL-T-5422. At the end of this test the equipment shall be subjected to and shall meet the requirements of the individual tests.

*4.8.33 Vibration Failure - The indicator shall be tested in accordance with Procedure 1, Part II, Vibration Test of MIL-T-5422, except the following shall be substituted in lieu of Curve II.a of Figure 6: Between 5 Hz and 20 Hz, a constant double amplitude of 0.10 inch shall be applied. Between 20 Hz and 2000 Hz, 2 gs shall be applied. Power, including power to the lighting circuit, rate-of-turn indicator, and flight director pointers, shall be applied to the unit during the vibration test. Following exposure to vibration, the unit shall meet the individual test requirements.

*4.8.34 Humidity - Subject the indicator to an atmosphere in which the relative humidity is maintained in excess of 95%, unless stated otherwise in the following steps. Moisture shall be provided by steam or by evaporation of water having a pH value between 6.5 and 7.5 measured at 25°C. The velocity of air throughout the exposure area shall not exceed 150 feet per minute. The test chamber shall be vented to the atmosphere to prevent the build-up of pressure. Provisions shall be made to prevent water from dripping onto the equipment above. The procedure shall be in accordance with the following steps:

Step 1 - Over a two-hour period, raise the chamber to 50 ± 3°C and increase the relative humidity to a value in excess of 95%.

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Step 2 - Maintain the chamber temperature at 50 ± 3°C with the relative humidity in excess of 95% for 6 hours.

Step 3 - During the next 16-hour period, decrease the temperature gradually to 38°C, or lower. During this period, keep the relative humidity as high as possible and do not allow it to fall below 85%.

Step 4 - Steps 1, 2, and 3 constitute a cycle. Repeat these steps until a total of 2 cycles (48 hours of exposure) have been completed.

Step 5 - At the end of the exposure period, remove the indicator from the humidity chamber and drain off (do not wipe) any condensed moisture. Place indicator properly connected, in a temperature chamber adjust for +71°C ± 2°C and apply power. At the end of 2 hours remove power and remove indicator from temperature chamber. Allow time to cool to room temperature (4 hours).

Step 6 - After the indicator has returned to room temperature it shall be subjected to and meet the requirements of the individual tests. There shall be no evidence of corrosion or other deterioration which will affect subsequent operation of the indicator.

*4.8.35 Salt Spray - The indicator shall be tested in accordance with Salt Spray Test of Specification MIL-T-5422. There shall be no damage to the indicator. At the end of the exposure period, the indicator shall meet the requirements of all the individual tests. No corrosion damage or malfunction of cage mechanism shall result from this test when in either caged or uncaged position.

*4.8.36 Fungus - The indicator shall be tested in accordance with Fungus Resistance Test of Specification MIL-T-5422, except that the test period shall be 14 days. No sticking or binding of the caging knob shall occur as a result of the test; there shall be no deterioration that would adversely affect the subsequent operation of the indicator, and fungus growth shall not be supported. At the end of the exposure period, the indicator shall meet the requirements of all the individual tests.

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*4.8.37 Sand and Dust Test - The equipment shall be subjected to Sand and Dust Tests, Specification MIL-T-5422. At the end of sand exposure the indicator shall not be damaged and shall meet the requirements of all individual tests.

*4.8.38 Radio Noise Interference - The indicator, properly connected, shall be tested for conducted and radiated radio noise interference in accordance with MIL-STD-461 and MIL-STD-462. Limits do not apply during switching of power failure flag by pulling caging knob.

*4.8.39 Temperature Shock - The indicator shall be subjected to Temperature Shock Test, of MIL-T-5422. At the conclusion of this test, the equipment shall be subjected to and shall meet the requirements of the individual tests.

4.8.40 Performance -

4.8.40.1 Longitudinal Accelerations - The equipment shall meet the requirements of 3.4.1.1.2 (see 4.8.6.2.3 for bench tests).

4.8.40.2 Coordinated Turns - The equipment shall meet the requirements of 3.4.1.1.3 (see 4.8.6.2.1 and 4.8.6.2.2 for bench tests).

4.8.40.3 Uncoordinated Turns - The equipment shall meet the requirements of 3.4.1.1.4 (by flight test by the procuring agency if required).

*4.8.41 Explosion Test - The indicator shall be tested in accordance with the Explosion Test of MIL-T-5422.

*4.8.42 Internal Examination - The case of the instrument may be opened and the internal mechanism examined for compliance with the Preproduction (First Article) requirements of the contract, detail specification, and all applicable specifications. Any deterioration or damage which could in any manner prevent the instrument from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

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4.9 Rejection and Retest - Equipment which has been rejected may be reworked to have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Government inspector.

*4.10 Maintainability Demonstration - Maintainability Demonstration Test shall be performed in accordance with MIL-STD-471 (if required by contract). If the Maintainability Demonstration Test has been performed on a previous contract or order, it need not be repeated.

4.11 Sealing and Adhesion - Where applicable, Elastomeric or resinous materials or combinations of these materials may be used as a cover glass seal provided that the specified sealing requirements are met under all tests and environmental conditions indicated herein. These materials shall be in accordance with MIL-S-8802 Adhesion materials for bonding must be in accordance with MIL-A-9067 and MMM-A-134. It is not intended to include seals accomplished by gaskets.

*5. PACKAGING

5.1 Packaging, Packing, Preservation, and Marking - Packaging shall be in accordance with MIL-C-81559.

6. NOTES

6.1 Intended Use - The 3-Inch Self-Contained Attitude Indicator is being intended for use in an aircraft as either a prime or standby vertical reference system. The indicator shall provide visual pitch and roll attitude as well as synchro type outputs indicating attitude. A flight director computer shall provide signals for operation of the vertical and horizontal pointers. The rate-of-turn indicator shall be controlled by a remote rate sensor.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(1) Title, number, and date of this specification.
(2) Selection of applicable levels of packaging and packing.

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6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This specification - This specification shall have precedence over all applicable subsidiary specification, any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Definitions -

6.5.1 Response Time - Response time is defined as the time required for the pointer to reach 90 percent of its final indication. The time is recorded from the instant of application of current. The pointer shall be considered at rest when it has reached within ± 1 percent of the scale length of the actual rest point.

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★ 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

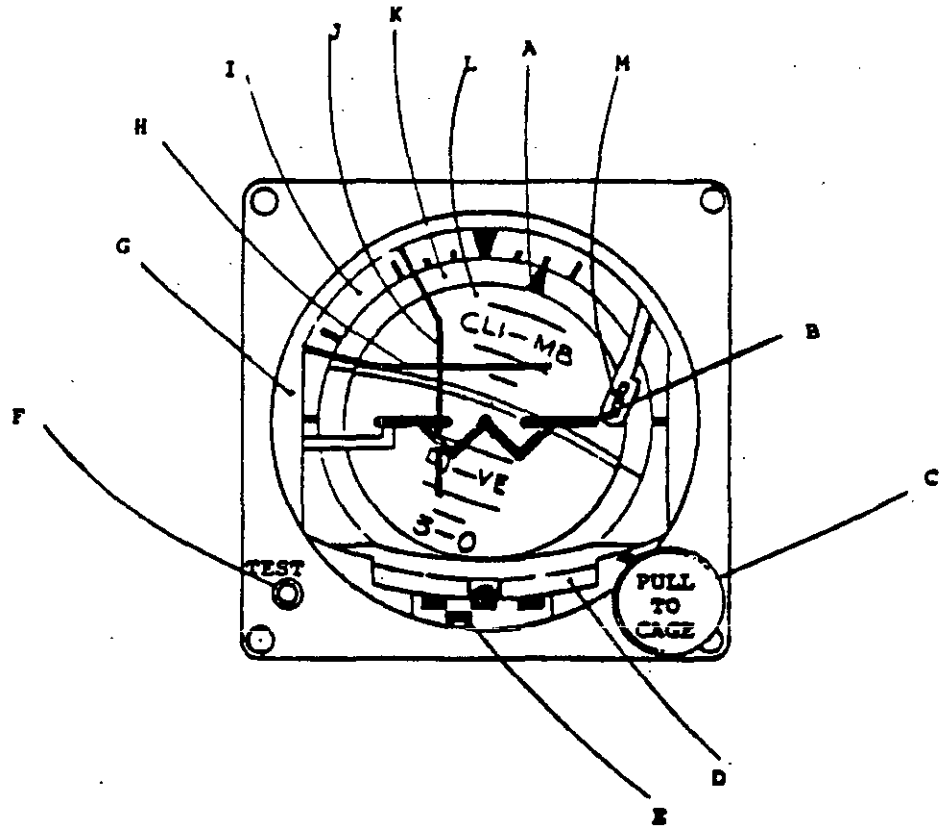
6.7 Jurisdiction - This specification is under the cognizance of AIR-53354C.

Custodian:
Navy-AS

Preparing Activity:
Navy-AS

Project No. 6610-NO94

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A. Bank Angle Index

B. Miniature Aircraft

C. Caging & Pitch Trim Knob

D. Slip Indicator Inclinometer

E. Rate-of-Turn Index Needle

F. Pointer-Deflecting Test Button

G. Pointer Shield

H. Horizontal Pointer Pitch Steering

I. Bank Scale

J. Vertical Pointer Bank Steering

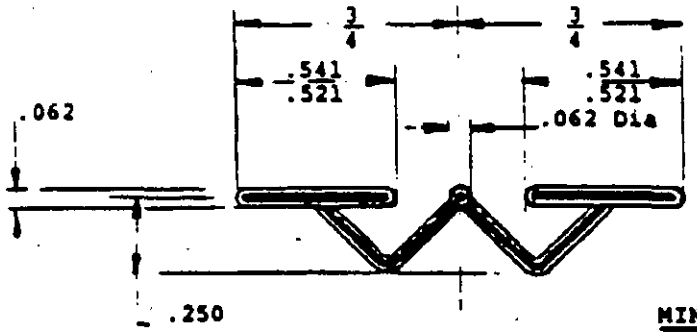
K. Mechanism Mask Sky-Pointer

L. Spheroid

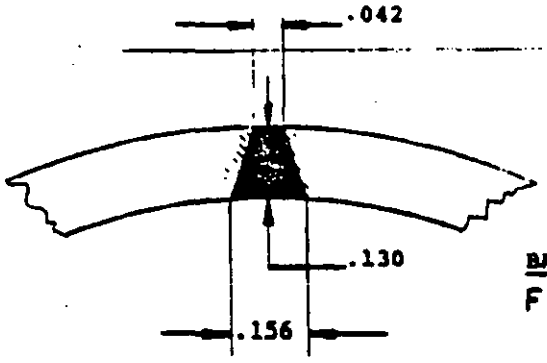
M. Power Failure Flag

FIGURE 1. INDICATOR DISPLAY.

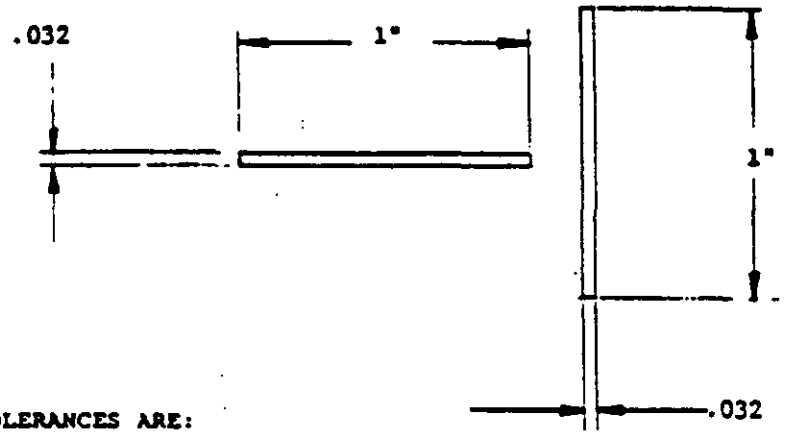
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MINIATURE AIRCRAFT.
FIGURE 2.



BANK INDEX.
FIGURE 3.



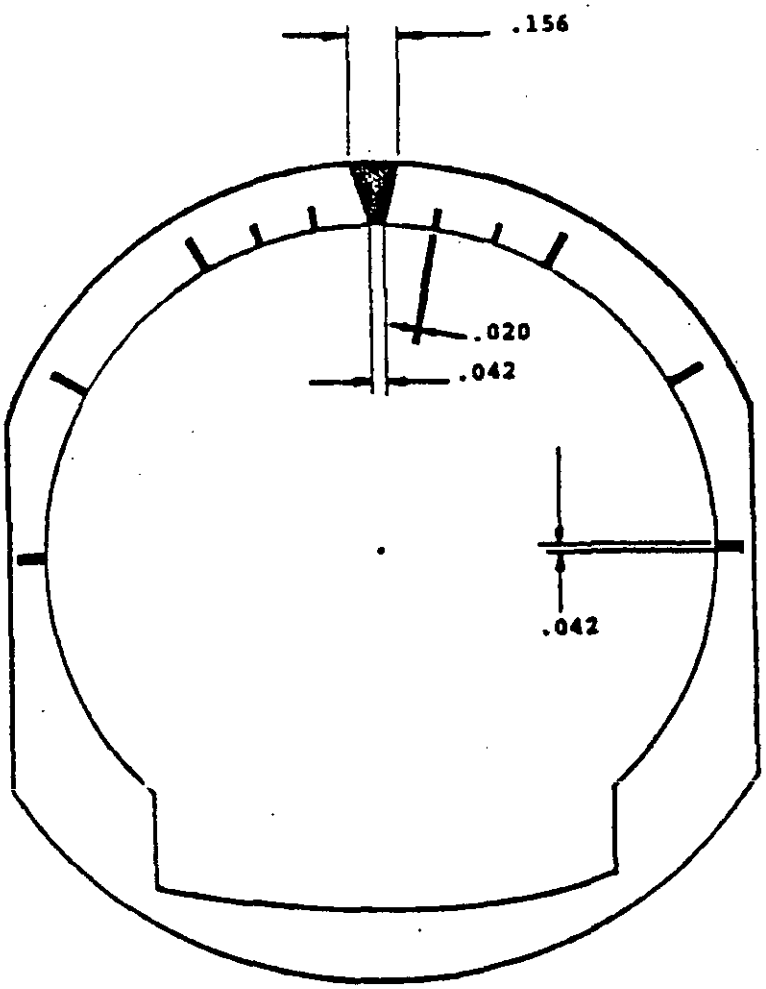
TOLERANCES ARE:

FRACTIONS ± 1/64
DECIMALS ± .005

FIGURE 4. VERTICAL & HORIZONTAL NEEDLES.

(Refer to Table V for Metric Conversion Chart)

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INDICES AT: ± 10°, & ± 20° ARE .062 LONG

INDICES AT: ± 30°, & ± 60° ARE .125 LONG

INDICES AT: ± 90°, ARE .100 LONG

TOLERANCES ARE:

FRACTIONS: ± 1/64
DECIMALS: ± .005

FIGURE 5. BANK ANGLE DIAL.

(Refer to Table V for Metric Conversion Chart)

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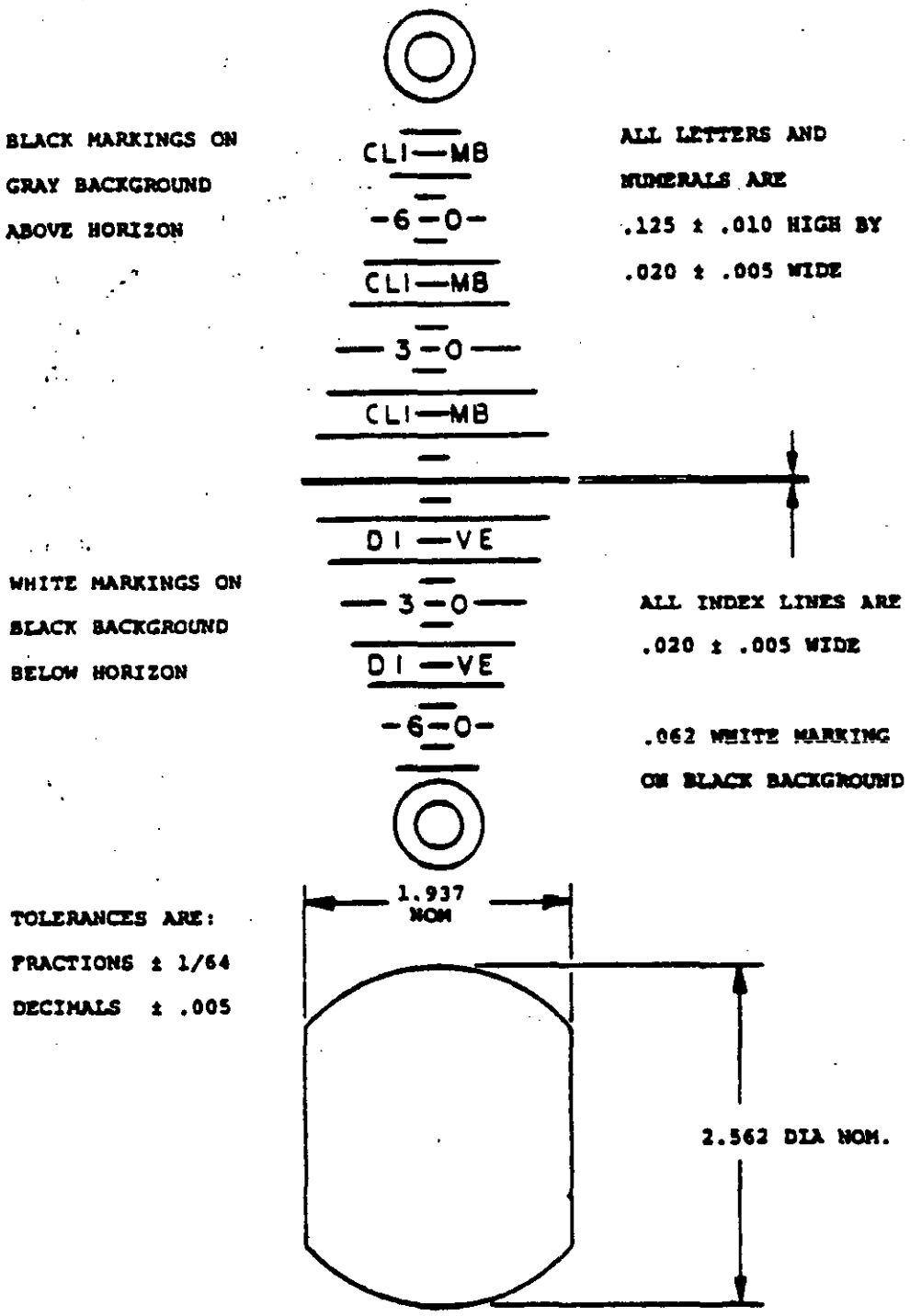
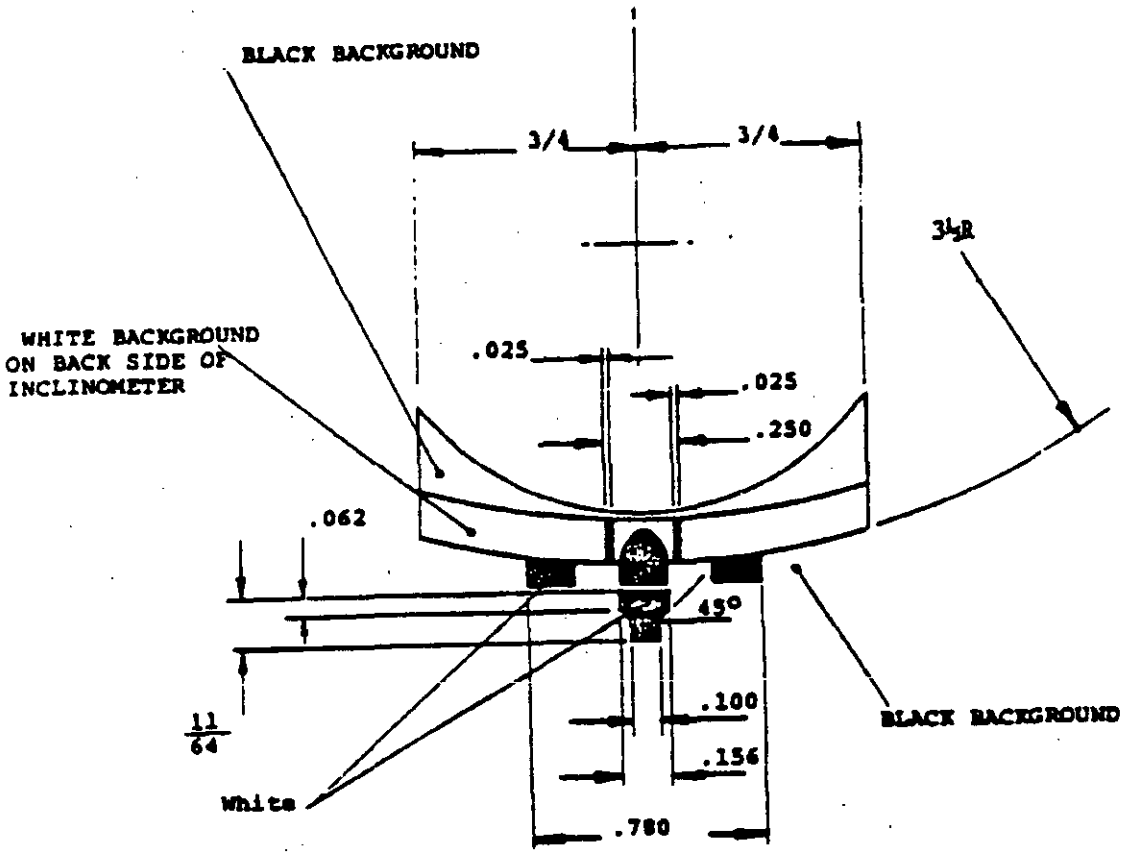


FIGURE 6. SPHEROID (ID-1791/A).

(Refer to Table V for Metric Conversion Chart)

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UNLESS OTHERWISE SPECIFIED:

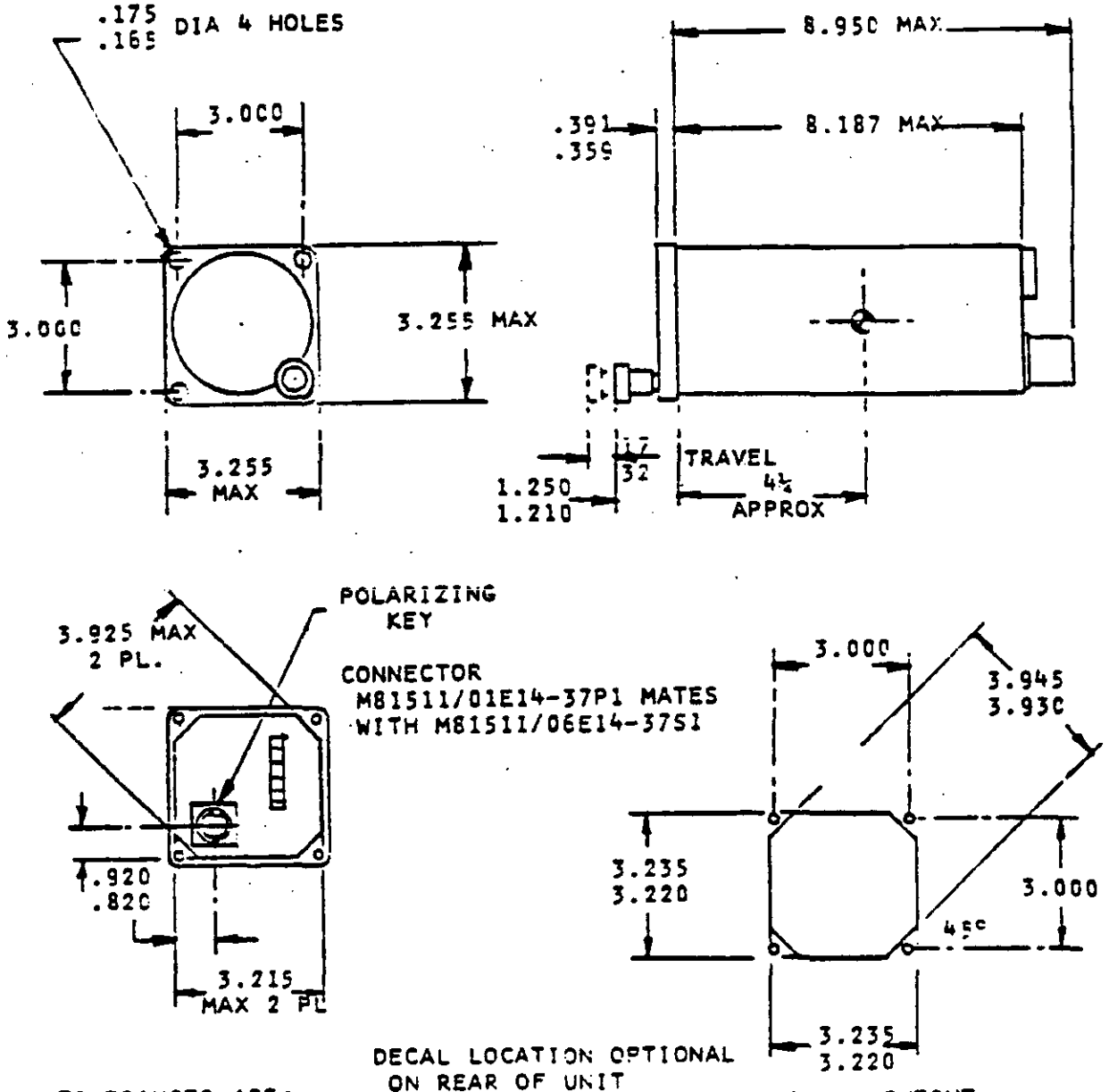
FRACTIONS	± 1/64
DECIMALS	± .005
ANGLES	± 1 Degree

INCLINOMETER AND RATE OF TURN.

FIGURE 7.

(Refer to Table V for Metric Conversion Chart)

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TOLERANCES ARE:
FRACTIONS: ±1/64
DECIMALS: ±.005
ANGLES: ±1°

FIGURE 8. OUTLINE DIMENSIONS.

(Refer to Table V for Metric Conversion Chart)

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TABLE V. Metric Conversion Chart.

FIGURE 2 MINIATURE AIRCRAFT, PAGE 56

Inches	MM
.021	1.57
.250	6.35
.521	13.23
.541	13.74
3/4	19.05

FIGURE 3 BANK INDEX, PAGE 56

Inches	MM
.042	1.07
.130	3.30
.156	3.96

FIGURE 4 VERTICAL AND HORIZONTAL NEEDLES, PAGE 56

Inches	MM
.032	.81
1.00	25.4

FIGURE 5 BANK ANGLE DIAL, PAGE 57

Inches	MM
.020	.51
.042	1.07
.062	1.57
.100	2.54
.125	3.17
.156	3.96

FIGURE 6 SPHEROID, PAGE 58

Inches	MM
.005	.13
.010	.25
.020	.51
.062	1.57
.125	3.17
1.937	49.20
2.562	65.07

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TABLE V (CONTINUED)

FIGURE 7 INCLINOMETER AND RATE-OF-TURN, PAGE 59

Inches	MM
.025	.63
.062	1.57
.100	2.54
.156	3.96
11/64	4.37
.250	6.35
3/4	19.05
.780	19.81
3-1/2	88.90

FIGURE 8 OUTLINE DIMENSIONS, PAGE 60

Inches	MM
.165	4.19
.175	4.44
.359	9.12
.391	9.93
17/32	13.50
.820	20.83
.920	23.37
1.210	30.73
1.250	31.75
3.000	76.20
3.215	81.66
3.220	81.79
3.235	82.17
3.255	82.68
3.925	99.69
3.930	99.82
3.945	100.20
4-1/4	107.95
8.187	207.95
8.950	227.33

NOTE: 1. METRIC EQUIVALENTS (TO THE NEAREST .01MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.

2. TOLERANCE CONVERSION FRACTIONS $\pm$ 1/64 = $\pm$ .4 MM
DECIMALS $\pm$.005 = $\pm$ .13 MM

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